OUTPUTS

EPOCH: 40

```
C:\Users\itz\Desktop>python neural_network.py
Training examples: 1437
Testing examples: 360
W1 shape: (64, 128)
W2 shape: (128, 10)
Predicted probabilities:
[[0.    0.    0.    0.    1.    0.    0.    0.    0.    0.   ]
 [0.    0.    0.    0.    1.    0.    0.    0.    0.    0.   ]
 [0.    0.    0.    0.    0.997 0.    0.    0.    0.003 0.   ]
 [0.    0.    0.    0.    0.975 0.    0.    0.    0.025 0.   ]
 [0.    0.    0.    0.    0.882 0.    0.117 0.    0.001 0.   ]]
Probability sums:
[1. 1. 1. 1. 1.]

C:\Users\itz\Desktop>python neural_network.py
Total images: 1797
Training images: 1437
Testing images: 360
Epoch 05 | Loss: 0.4694 | Train accuracy: 0.912 | Test accuracy: 0.900
Epoch 10 | Loss: 0.2549 | Train accuracy: 0.952 | Test accuracy: 0.931
Epoch 15 | Loss: 0.1839 | Train accuracy: 0.963 | Test accuracy: 0.939
Epoch 20 | Loss: 0.1483 | Train accuracy: 0.971 | Test accuracy: 0.942
Epoch 25 | Loss: 0.1260 | Train accuracy: 0.973 | Test accuracy: 0.958
Epoch 30 | Loss: 0.1093 | Train accuracy: 0.978 | Test accuracy: 0.961
Epoch 35 | Loss: 0.0956 | Train accuracy: 0.979 | Test accuracy: 0.956
Epoch 40 | Loss: 0.0902 | Train accuracy: 0.985 | Test accuracy: 0.964

Final test accuracy: 0.9638888888888889
```

EPOCH: 80

```
C:\Users\itz\Desktop>python neural_network.py
Total images: 1797
Training images: 1437
Testing images: 360
Epoch 05 | Loss: 0.4694 | Train accuracy: 0.912 | Test accuracy: 0.900
Epoch 10 | Loss: 0.2549 | Train accuracy: 0.952 | Test accuracy: 0.931
Epoch 15 | Loss: 0.1839 | Train accuracy: 0.963 | Test accuracy: 0.939
Epoch 20 | Loss: 0.1483 | Train accuracy: 0.971 | Test accuracy: 0.942
Epoch 25 | Loss: 0.1260 | Train accuracy: 0.973 | Test accuracy: 0.958
Epoch 30 | Loss: 0.1093 | Train accuracy: 0.978 | Test accuracy: 0.961
Epoch 35 | Loss: 0.0956 | Train accuracy: 0.979 | Test accuracy: 0.956
Epoch 40 | Loss: 0.0902 | Train accuracy: 0.985 | Test accuracy: 0.964

Final test accuracy: 0.9638888888888889
Model saved to digit_model.npz

C:\Users\itz\Desktop>python predict_digit.py
Predicted digit: 5
Model confidence: 47.77%

Probabilities:
0:   4.16%
1:  12.01%
2:   0.48%
3:   4.51%
4:   2.25%
5:  47.77%
6:   1.63%
7:   2.45%
8:   2.73%
9:  22.04%
```

```
C:\Users\itz\Desktop>python predict_digit.py
Predicted digit: 5
Model confidence: 47.77%

Probabilities:
0:   4.16%
1:  12.01%
2:   0.48%
3:   4.51%
4:   2.25%
5:  47.77%
6:   1.63%
7:   2.45%
8:   2.73%
9:  22.04%

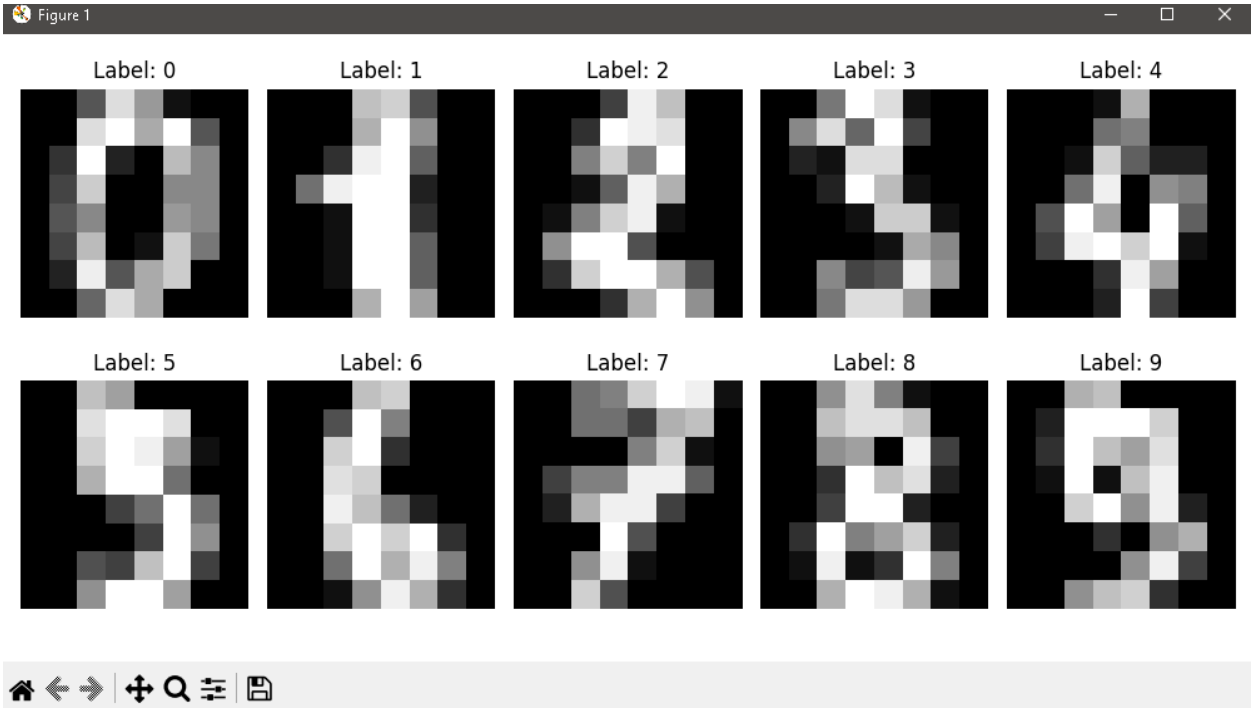
C:\Users\itz\Desktop>python neural_network.py
Total images: 1797
Training images: 1437
Testing images: 360
Epoch 05 | Loss: 0.4694 | Train accuracy: 0.912 | Test accuracy: 0.900
Epoch 10 | Loss: 0.2549 | Train accuracy: 0.952 | Test accuracy: 0.931
Epoch 15 | Loss: 0.1839 | Train accuracy: 0.963 | Test accuracy: 0.939
Epoch 20 | Loss: 0.1483 | Train accuracy: 0.971 | Test accuracy: 0.942
Epoch 25 | Loss: 0.1260 | Train accuracy: 0.973 | Test accuracy: 0.958
Epoch 30 | Loss: 0.1093 | Train accuracy: 0.978 | Test accuracy: 0.961
Epoch 35 | Loss: 0.0956 | Train accuracy: 0.979 | Test accuracy: 0.956
Epoch 40 | Loss: 0.0902 | Train accuracy: 0.985 | Test accuracy: 0.964
Epoch 45 | Loss: 0.0814 | Train accuracy: 0.985 | Test accuracy: 0.964
Epoch 50 | Loss: 0.0727 | Train accuracy: 0.987 | Test accuracy: 0.961
Epoch 55 | Loss: 0.0659 | Train accuracy: 0.990 | Test accuracy: 0.964
Epoch 60 | Loss: 0.0624 | Train accuracy: 0.991 | Test accuracy: 0.964
Epoch 65 | Loss: 0.0589 | Train accuracy: 0.992 | Test accuracy: 0.967
Epoch 70 | Loss: 0.0534 | Train accuracy: 0.993 | Test accuracy: 0.972
Epoch 75 | Loss: 0.0513 | Train accuracy: 0.994 | Test accuracy: 0.964
Epoch 80 | Loss: 0.0481 | Train accuracy: 0.996 | Test accuracy: 0.972

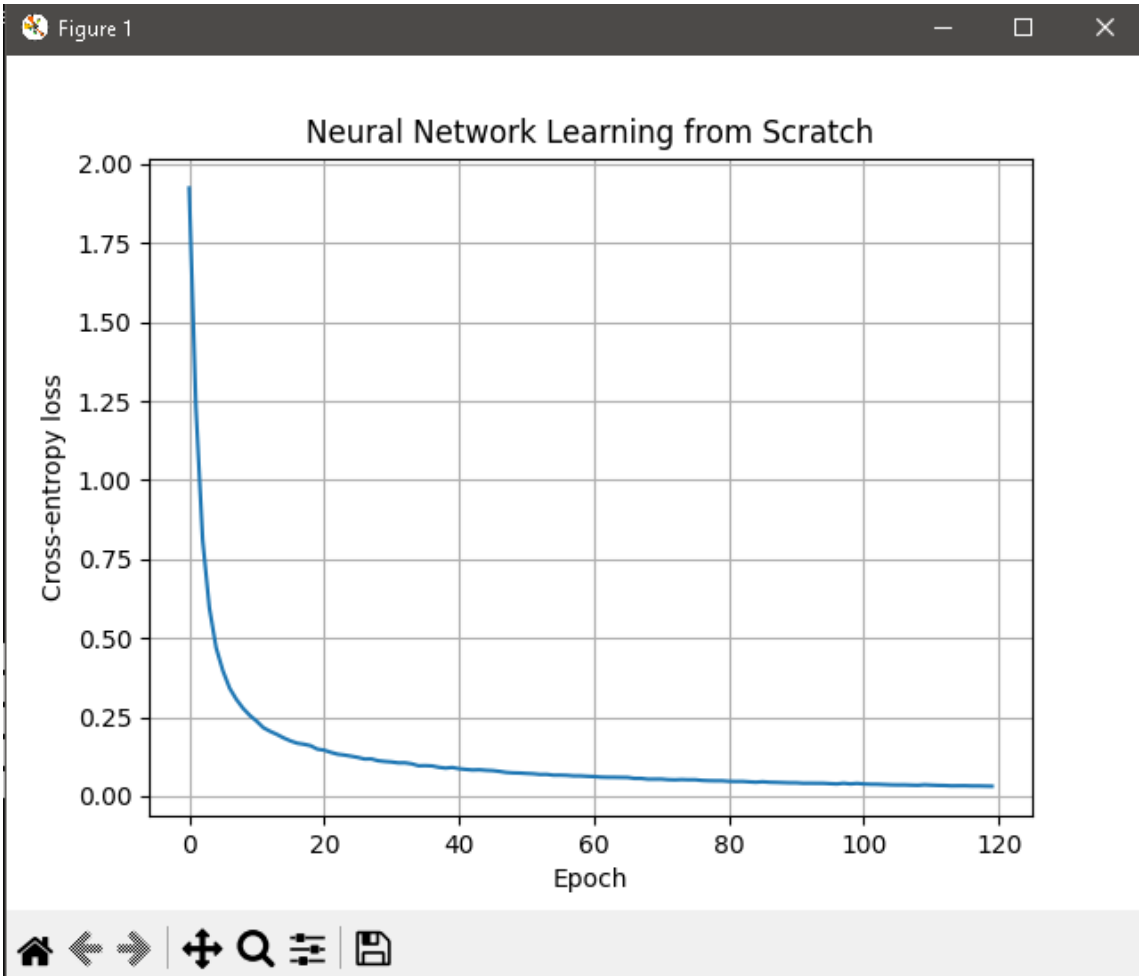
Final test accuracy: 0.9722222222222222
Model saved to digit_model.npz
```

EPOCH: 120

```
C:\Users\itz\Desktop>python neural_network.py
Total images: 1797
Training images: 1437
Testing images: 360
Epoch 05 | Loss: 0.4694 | Train accuracy: 0.912 | Test accuracy: 0.900
Epoch 10 | Loss: 0.2549 | Train accuracy: 0.952 | Test accuracy: 0.931
Epoch 15 | Loss: 0.1839 | Train accuracy: 0.963 | Test accuracy: 0.939
Epoch 20 | Loss: 0.1483 | Train accuracy: 0.971 | Test accuracy: 0.942
Epoch 25 | Loss: 0.1260 | Train accuracy: 0.973 | Test accuracy: 0.958
Epoch 30 | Loss: 0.1093 | Train accuracy: 0.978 | Test accuracy: 0.961
Epoch 35 | Loss: 0.0956 | Train accuracy: 0.979 | Test accuracy: 0.956
Epoch 40 | Loss: 0.0902 | Train accuracy: 0.985 | Test accuracy: 0.964
Epoch 45 | Loss: 0.0814 | Train accuracy: 0.985 | Test accuracy: 0.964
Epoch 50 | Loss: 0.0727 | Train accuracy: 0.987 | Test accuracy: 0.961
Epoch 55 | Loss: 0.0659 | Train accuracy: 0.990 | Test accuracy: 0.964
Epoch 60 | Loss: 0.0624 | Train accuracy: 0.991 | Test accuracy: 0.964
Epoch 65 | Loss: 0.0589 | Train accuracy: 0.992 | Test accuracy: 0.967
Epoch 70 | Loss: 0.0534 | Train accuracy: 0.993 | Test accuracy: 0.972
Epoch 75 | Loss: 0.0513 | Train accuracy: 0.994 | Test accuracy: 0.964
Epoch 80 | Loss: 0.0481 | Train accuracy: 0.996 | Test accuracy: 0.972
Epoch 85 | Loss: 0.0436 | Train accuracy: 0.994 | Test accuracy: 0.964
Epoch 90 | Loss: 0.0417 | Train accuracy: 0.995 | Test accuracy: 0.972
Epoch 95 | Loss: 0.0402 | Train accuracy: 0.994 | Test accuracy: 0.969
Epoch 100 | Loss: 0.0398 | Train accuracy: 0.996 | Test accuracy: 0.967
Epoch 105 | Loss: 0.0354 | Train accuracy: 0.997 | Test accuracy: 0.972
Epoch 110 | Loss: 0.0354 | Train accuracy: 0.997 | Test accuracy: 0.961
Epoch 115 | Loss: 0.0324 | Train accuracy: 0.999 | Test accuracy: 0.972
Epoch 120 | Loss: 0.0309 | Train accuracy: 0.998 | Test accuracy: 0.972

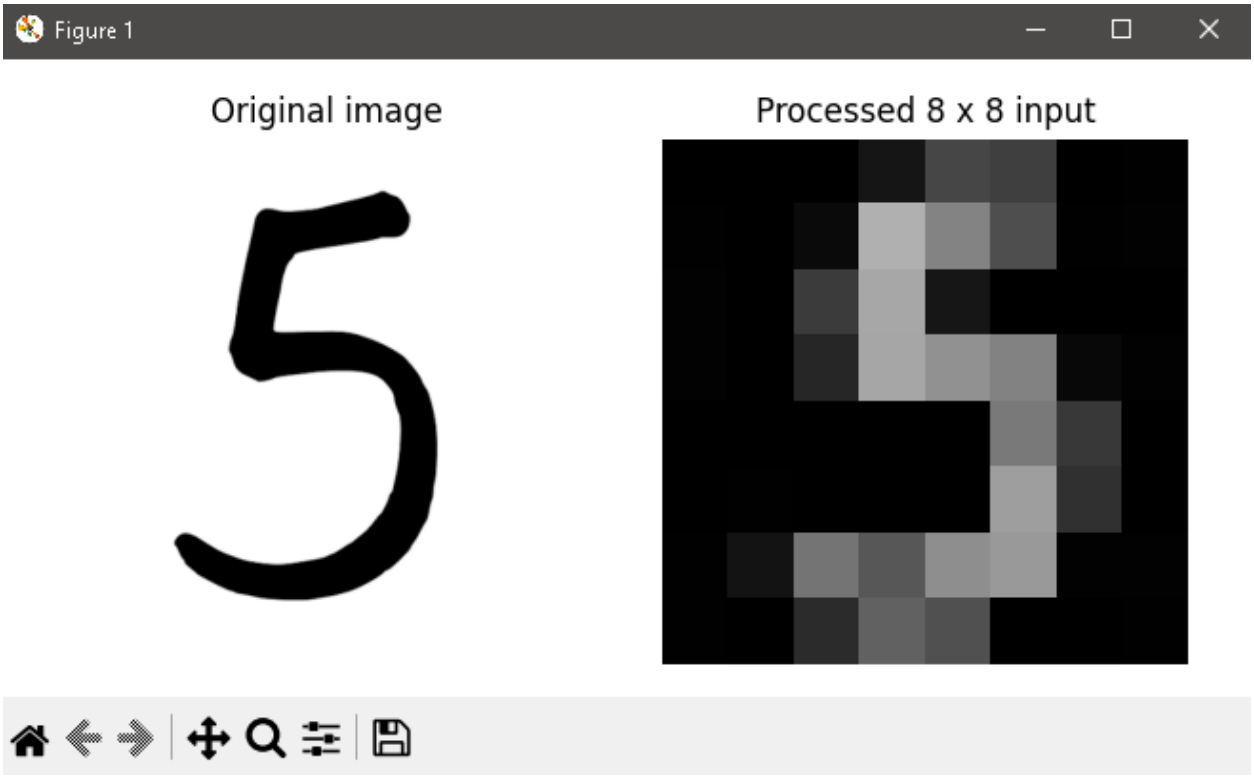
Final test accuracy: 0.9722222222222222
Model saved to digit_model.npz
```


Figure 1
Label: 0
Label: 1
Label: 2
Label: 3
Label: 4
Label: 5
Label: 6
Label: 7
Label: 8
Label: 9


Figure 1
Neural Network Learning from Scratch
Cross-entropy loss
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
20
40
60
80
100
120
Epoch

```
C:\Users\itz\Desktop>python predict_digit.py
Predicted digit: 5
Model confidence: 75.79%

Probabilities:
0:   2.80%
1:   7.01%
2:   0.10%
3:   2.27%
4:   0.76%
5:  75.79%
6:   1.03%
7:   1.20%
8:   0.94%
9:   8.09%
```



ANALYSIS BY AI

